PHY 456 - Tutorial 7
Addition of Angular Momentum

Qu: what are the states of the two particle system in the Hilbert space $\frac{1}{2} \otimes \frac{1}{2}$?

$$S^2 = (S_1 + S_2)^2 \quad \text{and}$$

$$S_z = S_{z1} + S_{z2}$$

can be simultaneously diagonalized.

This therefore a 4-dimensional Hilbert space with basis kets

$$|++\rangle \qquad |-+\rangle$$

$$|+-\rangle \qquad |--\rangle$$

which can diagonalize S^2, S_z:

$$S_z|++\rangle = (S_{z1} + S_{z2})|++\rangle = \hbar|++\rangle$$

$$S_z|+-\rangle = \ldots = S_z|-+\rangle = 0$$

$$S_z |--\rangle = \dots = -\hbar |--\rangle.$$

or the matrix representation $S_z = \hbar \begin{pmatrix} 1 & & & 0 \\ & 0 & & \\ & & 0 & \\ 0 & & & -1 \end{pmatrix}$

which is already diagonal. Note two degenerate eigenvalues with eigenkets $|+-\rangle, |-+\rangle$.

Similarly for $S^2 = (S_1 + S_2)\cdot(S_1 + S_2)$

$$= S_1^2 + S_2^2 + 2 S_1 \cdot S_2$$

$$+ 2(S_{1x} S_{2x} + S_{1x} S_{2y} + S_{1z} S_{2z})$$

Recall raising and lowering operators for spin-$\frac{1}{2}$

$$S_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

$$S_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

expanded into the pauli matrices

$$\boxed{S_\pm = \frac{1}{2}(S_x \pm i S_y)}$$

Then $S_{+_1} S_{-_2} = (S_{x1} + i S_{y1})(S_{x2} - i S_{y2})$

$$= S_{x1} S_{x2} - i S_{x1} S_{y2} + i S_{y1} S_{x2} + S_{y1} S_{y2}$$

Similarly $S_{-1}S_{+2} = S_{x1}S_{x2} + iS_{x1}S_{y2} - iS_{y1}S_{x2} + S_{y1}S_{y2}$

So adding these reduces the dot product into

$$2\, S_1 \cdot S_2 = S_{+1}S_{-2} + S_{-1}S_{+2} + 2S_{1z}S_{2z}$$

and

$$S^2 = S_1^2 + S_2^2 + S_{+1}S_{-2} + S_{-1}S_{+2} + 2S_{1z}S_{2z}.$$

Acting on $S^2|++\rangle = \hbar^2\left(\frac{3}{4} + \frac{3}{4} + 0 + 0 + \frac{2}{4}\right)|++\rangle$

$= 2\hbar^2|++\rangle$ already an eigenstate

Similarly $S^2|\pm\mp\rangle = \hbar^2\left(\frac{3}{4} + \frac{3}{4} \left\{\begin{matrix} +0 & +|-+\rangle \\ +|+-\rangle & +0 \end{matrix}\right\} - \frac{2}{4}\right)|\pm\mp\rangle$

$= \hbar^2|\pm\mp\rangle$

$S^2|--\rangle = 2\hbar^2|--\rangle.$

So

$$\begin{array}{cc} & \begin{array}{cccc} |++\rangle & |+-\rangle & |-+\rangle & |--\rangle \end{array} \\ \hbar^2 & \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 \end{pmatrix} \end{array}$$

is the S^2 matrix representation of the S^2 operator for the $\frac{1}{2} \otimes \frac{1}{2}$ diagonalization.

Observe that the $M_{1,4 \times 1,4} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ is the matrix minor which can be expanded as

$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= \mathbb{I} + \sigma_x$$

whose eigenstates are $\frac{1}{\sqrt{2}}\left(|+-\rangle \pm |-+\rangle \right)$

with eigenvalues $2 \, (\hbar^2)$ and $0 \, (\hbar^2)$

$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 1 \end{pmatrix}$; $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} 1 \\ -1 \end{pmatrix} = 0\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

The diagonal form is therefore

$$S^2_{diag} = \begin{pmatrix} 2 & & & \\ & 2 & & \\ & & 2 & \\ & & & 0 \end{pmatrix}$$

with eigenvalues $\underbrace{2\hbar^2}_{1(1+1)}, \underbrace{0\hbar^2}_{0(0+1)}$

and $2\hbar^2$ is threefold degenerate.